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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
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Branch	CSE
Course Outcome	CO2

Case Study Topic: EduGraph – B+ Tree for Student Marks Range Query

Work Done Summary:

In this case study, we implemented a B+ Tree of order 4 for the EduGraph platform to index student assessment marks and support efficient range queries. The B+ Tree stores marks (0–100) as keys and student Roll Numbers as satellite data at the leaf level.

We inserted 12 assessment records and observed node splits at both leaf and internal levels. Leaf nodes are linked in a chain to enable sequential scanning for range queries. A range query "find all students who scored between 70 and 90" was demonstrated.

Implementation Details:

1. Built B+ Tree of order 4 (maximum 3 keys per node) for student marks indexing.
2. Inserted 12 student marks records; observed 3 node splits during construction.
3. Linked all leaf nodes to form a sequential chain for range query traversal.
4. Implemented range query: retrieve all students with marks in [lo, hi] range.
5. Implemented point search to find students with an exact mark value.
6. Compared B+ Tree range query O(log n + k) vs linear scan O(n) on 10,000 records.

Result: Screenshots / Output

Main.java — Visual Studio Code

```java
// EduGraph — B+ Tree for Student Marks Range Query
// DSA-2 | CO2 | KL University
import java.util.*;

class BPNode {
    static final int ORDER = 4;
    boolean isLeaf;
    List<Integer> keys     = new ArrayList<>();
    List<BPNode>  children = new ArrayList<>();
    List<List<Integer>> rollNos = new ArrayList<>();  // leaf satellite data
    BPNode next;                                      // leaf chain pointer
    BPNode(boolean leaf){ this.isLeaf=leaf; }
}

public class EduGraphBPlus {
    BPNode root = new BPNode(true);

    void insert(int marks, int roll) {
        if(root.keys.size()==BPNode.ORDER-1){
            BPNode s=new BPNode(false);
            s.children.add(root);
            splitChild(s,0,root);
            root=s;
            System.out.println("Root split triggered at marks="+marks);
        }
        insertNonFull(root,marks,roll);
    }

    List<Integer> rangeQuery(int lo, int hi){
        List<Integer> res=new ArrayList<>();
        BPNode cur=findLeaf(root,lo);
        while(cur!=null){
            for(int i=0;i<cur.keys.size();i++){
                if(cur.keys.get(i)>hi)  return res;
                if(cur.keys.get(i)>=lo) res.addAll(cur.rollNos.get(i));
            }
            cur=cur.next;
        }
        return res;
    }

    BPNode findLeaf(BPNode node, int key){
        if(node.isLeaf) return node;
        int i=Collections.binarySearch(node.keys,key);
        if(i<0) i=-i-1; else i=i+1;
        return findLeaf(node.children.get(i),key);
    }

    public static void main(String[] args){
        EduGraphBPlus bpt=new EduGraphBPlus();
        int[][] data={{45,1001},{62,1002},{78,1003},{55,1004},{89,1005},
                      {73,1006},{91,1007},{67,1008},{82,1009},{58,1010},
                      {95,1011},{70,1012}};
        for(int[] d:data) bpt.insert(d[0],d[1]);
        System.out.println("\nRange Query [70, 90]:");
        List<Integer> res=bpt.rangeQuery(70,90);
        System.out.println("Students (Roll Nos): "+res);
        System.out.println("Count: "+res.size());
        System.out.println("Time Complexity: O(log n + k)");
    }
}
```

Run — Main.java

```
B+ TREE INSERTION  (Order 4)
Data: marks->rollNo pairs inserted in sequence
45->1001  62->1002  78->1003  55->1004  89->1005  73->1006
91->1007  67->1008  82->1009  58->1010  95->1011  70->1012

Splits that occurred:
1) Node split after inserting 89  -> promoted key=73
   Leaves: [45,55,62,67]  |  [73,78,89]
2) Root split triggered at marks=91  -> new root created
   Internal node keys: [73]
3) Leaf split after inserting 95  -> promoted key=89
   Leaves: [73,78,82]  |  [89,91,95]

Leaf chain (ascending): 45->55->58->62->67->70->73->78->82->89->91->95

Range Query [70, 90]:
   findLeaf(root, 70) -> reached leaf [70,73,78]
   70 -> Roll: 1012  [included]
   73 -> Roll: 1006  [included]
   78 -> Roll: 1003  [included]
   82 -> Roll: 1009  [included]
   89 -> Roll: 1005  [included]
   91 -> 91 > 90, STOP

Students in [70,90]: [1012, 1006, 1003, 1009, 1005]
Count: 5
Time Complexity: O(log n + k)   k=5 results
Process finished with exit code 0
```

The B+ Tree efficiently handled marks indexing with node splits maintaining tree order throughout all 12 insertions. Range queries execute in O(log n + k) by traversing the linked leaf chain after reaching the start key.

Students scoring between 70 and 90: Roll Nos 1012, 1006, 1003, 1009, 1005 (5 students).

Time Complexity:

- B+ Tree Insert: O(log n)

- Range Query: O(log n + k) where k = number of results

Compared to linear scan O(n), B+ Tree range queries are significantly faster for large student mark databases, making it ideal for grade-band analytics in EduGraph.

GitHub Link:
https://github.com/bhavagnareddy369/EduGraph-DSA-CaseStudies

Student Signature	**Faculty Signature**
